DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha
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Research Interests

My research explores computational design and creativity, with a focus on representing and supporting creative intent in generative and interactive systems. I build models and interfaces that translate high-level human intent into expressive, controllable outcomes, spanning domains such as computer graphics, motion synthesis, and kinematic design.

Education

Georgia Tech, Ph.D. in Computer Science *2022–Present*

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE *2020–2022*

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE *2012–2020*

Experience

Ph.D. Research Intern, Dolby Laboratories *2024–2025*

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. *2023*

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. *2021–2022*

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador *2020–2022*

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. *2019*

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) *2017–2019*

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, "MotionSmith: A Sketch-Based Design System for Automata Making," *CHI*, Apr. 2026.
2 J. Yu, P. Garg, **D. Synn**, and H. Oh, "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning," *CHI*, Apr. 2025.